

Functional validation of novel HCC biomarkers identified by artificial intelligence and machine learning approach

Hepatocellular carcinoma (HCC) currently ranks as the sixth most frequently diagnosed cancer and the third highest cause of cancer-related deaths worldwide, causing about 830,000 deaths.[1] Detection of HCC in its early stages remains challenging due to a lack of clinical symptoms and specific biomarkers. Thus, the majority of tumors are detected at the late stages when curative treatment options are very limited. Development of HCC involves multistep process, where hepatocytes or hepatic progenitor cells proliferate in the setting of chronically altered and inflammatory microenvironment. Aberrant intracellular signaling and dysregulated genetic and epigenetic events, induced by constant cellular damage and turnover, subsequently progress into dysplastic nodules (low grade - LGDN and high grade - HGDN) and ultimately lead to formation of early and advanced HCC (eHCC and pHCC).[2] Liver cirrhosis is the strongest and most common risk factor for development of HCC.[3] More than 85% of HCC develop on the basis of advanced hepatic fibrosis and/or cirrhosis. The cumulative risk of HCC development is significantly influenced by the underlying etiological condition.[4] Several different risk factors have been associated with liver cirrhosis and HCC. The most prominent are chronic HBV and HCV infections, alcoholic liver disease, NAFLD/NASH, aflatoxin B1, metabolic disorders (e.g. hereditary haemochromatosis, α 1 antitrypsin deficiency, hereditary tyrosinemia), diabetes mellitus and several others with lower frequency (Figure 5).[5]

Recently, artificial intelligence (AI) and machine learning (ML) algorithms, as well as other statistical models, have played a critical role in various areas of human health and disease. These approaches also offer promising applications for the diagnosis and treatment of human cancers, which could facilitate the improvement of patient outcomes.[6]

The first part of this project focused on the application of artificial intelligence and machine learning algorithms using multiple transcriptomic datasets to identify novel biomarkers for the detection of early-stage HCC as well as to identify the relevant genes that are dysregulated during the progression from early to late stage. For this purpose, the HCC dataset consisting of a total of 770 samples with five subclasses representing the tissue samples of the patients was used:

- Surrounding liver (SL)
- Low-grade dysplastic nodules (LGDN)
- High-grade dysplastic nodules (HGDN)
- Early hepatocellular carcinoma (eHCC)
- Advanced hepatocellular carcinoma (pHCC)

The data has been obtained from three datasets including:

1. Lübeck dataset (AG Marquardt)
2. TCGA
3. GSE89377

First, the dataset was created using a semi-supervised machine learning approach. Samples from the TCGE and GSE89377 dataset were added to the Lübeck and labelled according to its class labels. All the genes were derived using AI-based algorithms, the K_best combined with mutual_info algorithm, to select the best features (Table 1). The samples distributed among the five labels were further reduced to two labels (SL, LGDN and HGDN) as one pre-malignant group and (eHCC and pHCC) as the second malignant HCC group (Figure 1 top panel). Statistical ANOVA test was used to evaluate the significance of the differences in gene expression levels among the different labels.

From the obtained data (Table 1), for further functional validation were selected two significant genes (OTUB1 and DNAJB14, Figure 1 top panel) that were not previously reported to be involved in HCC. OTUB1 is a deubiquitinating enzyme that plays a crucial role in various cellular processes by removing ubiquitin from specific protein substrates. This regulation affects protein stability, localization, and activity. There is growing evidence that OTUB1 regulates many cancer-related signaling pathways, including MAPK, ERα, epithelial-mesenchymal transition (EMT), RHOa, mTORC1, FOXM1 and P53, to promote tumor cell survival, proliferation, invasiveness and therapeutic resistance. In addition, clinical

Gene Name	P- value
SELENBP1	2.780759500652043e-19
MASP1	1.402621104026893e-38
PPP1R14B	2.69789539031622e-24
TAF2	1.3429424730931303e-33
OTUB1	1.6504164943816253e-11
HGF	5.694924453489817e-63
PDGFRA	1.083474507781217e-55
HIF1AN	6.750136105744364e-25
SAE1	2.339722474642119e-21
IFNA5	6.652562375434264e-08
IVD	9.723135905151953e-69
DNAJB14	4.5325842861879285e-111
ITGB8	2.9317356193968864e-69
PADI4	1.354242941604978e-77

Table 1. Gene candidates identified by AI/ML approach. Potential biomarkers involved in hepatocarcinogenesis discovered based on transcriptomic profiles of non-malignant and malignant HCC samples using AI/ML approaches.

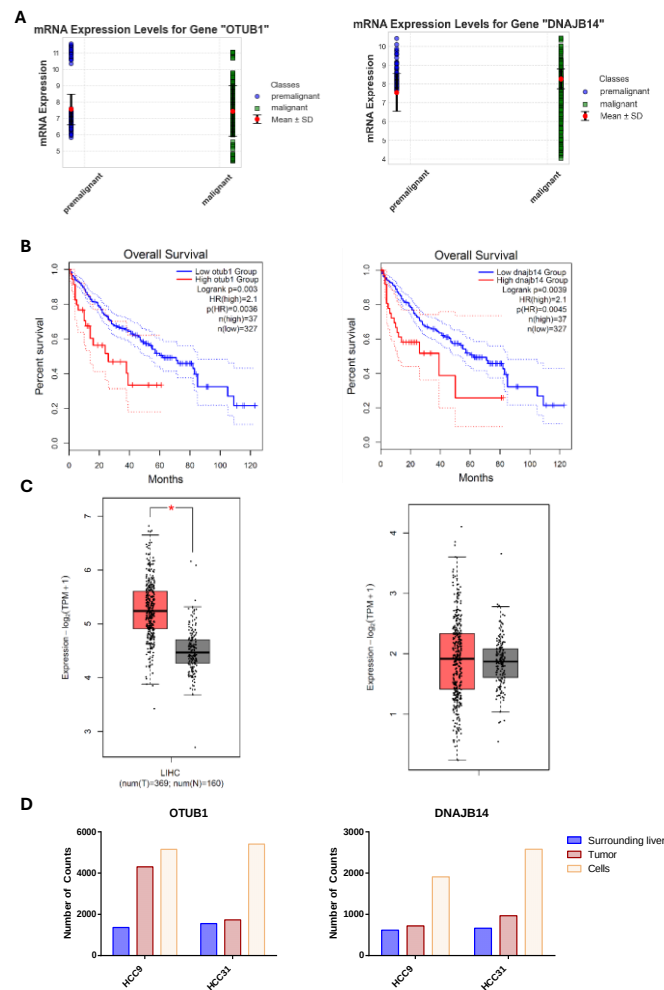


Figure 1. Expression levels of candidate genes and overall survival. A) Expression levels of OTUB1 and DNAJB14 obtained from three datasets (Lübeck, TCGA, GSE89377) after integration using AI/ML B) Panel showing overall survival of the patients based on expression of the target genes from the public dataset (TCGA); C) Expression level of OTUB1 and DNAJB14 relative to normal control obtained from the public dataset (TCGA). D) Gene expression level represented as normalized count number in tumor surrounding liver, core tumor tissue and corresponding primary patient-derived cell lines.

studies have associated elevated OTUB1 expression with high grade, invasiveness and metastasis in several tumor types, including lung, breast, ovarian, glioma, colon and gastric.[7] DNAJB14 is a member of the HSP40 family that functions as a co-chaperone with HSP70/Hsc70. It is involved in protein folding, trafficking and prevention of protein aggregation. DNAJB4 may be involved in the regulation of invasion and migration of breast cancer (BC) cells, and its expression level is associated with the prognosis of BC patients. Therefore, studies of DNAJB4 in other types of cancer, including HCC, may prove to be a new diagnostic/treatment biomarker.[8] Analysis of publicly available HCC datasets showed that patients with high expression of both genes had significantly lower overall survival than those with low expression, demonstrating their importance in HCC (Figure 1 B). The level of expression in HCC compared to normal liver showed significant upregulation of OTUB1, while DNAJB14 did not reach significant levels (Figure 1 C). Nevertheless, the expression in the primary HCC cell lines and corresponding tumor tissue, demonstrate higher level than in surrounding liver tissue. Together, the presented evidence speaks in favor of utilization of both biomarkers in HCC.

Objectives:

- ❖ Perform molecular and functional validation of the gene candidates in primary and established HCC cell lines.
- ❖ Validation in human tissue samples (in case of enough time)

Tasks:

1. Investigate the expression of OTUB1 and DNAJB14 in primary/established HCC cell lines and compare with the expression in immortalized hepatocytes
 - a. qRT-PCR
 - i. Primer design and order
 - ii. RNA isolation
 - iii. Run the analysis
 - b. Western blot
 - i. Isolate proteins
 - ii. Run WB
2. Select the cell lines with high vs low expression of the targets and perform siRNA mediated knockdown (KD)
 - a. siRNA KD
 - i. Define optimal conditions for siRNA KD
 - ii. Perform KD
 - iii. Validate by running qRT-PCR and WB
3. Functional validation (knockdown/chemical inhibition)
 - a. Chemical inhibition of OTUB1
 - i. Dose-response analysis
 - ii. Define IC50, IC25, IC10 concentrations
 - iii. Treat the cells with defined concentration and perform colony and sphere formation assay
 - b. Gene KD
 - i. Viability
 - ii. Colony and sphere formation
4. Validation
 - a. Stain by IHC/perform WB for the expression of the candidates on human tissue samples (normal liver and HCC)

Cell lines to be assessed:

THLE5B – control cells; immortalized hepatocytes

HCC9, HCC31 – primary patient-derived HCC cell lines

Huh7, HepG2 – commercial HCC cell lines

From the given cell lines, qRT-PCR and Western blot analyses were performed to assess the gene and protein levels of OTUB1¹ and DNAJB14 and normalized to control cells (THLE-5B). The results showed that among the cell lines, OTUB1 gene expression was upregulated in all cell lines compared to the control. DNAJB14 showed upregulation in Huh7 cells, while HepG2 was similar to THLE-5B². On the other hand, primary cell lines (HCC9 and HCC31) showed downregulation. To validate whether the transcribed gene levels correspond to the protein levels, WB analysis showed that the highest signal of DNAJB14 was present in the HepG2 and Huh7 cell lines, while primary cells had significantly lower levels. As the control cell lines do not fully represent normal hepatocytes (immortalized cells), relatively high levels of DNAJB14 are also expected. We are currently investigating the levels of this protein in human liver samples, which will serve as a control. Based on the RNA sequencing results (Figure 1D), it is expected that levels will be significantly higher in HCC cell lines.

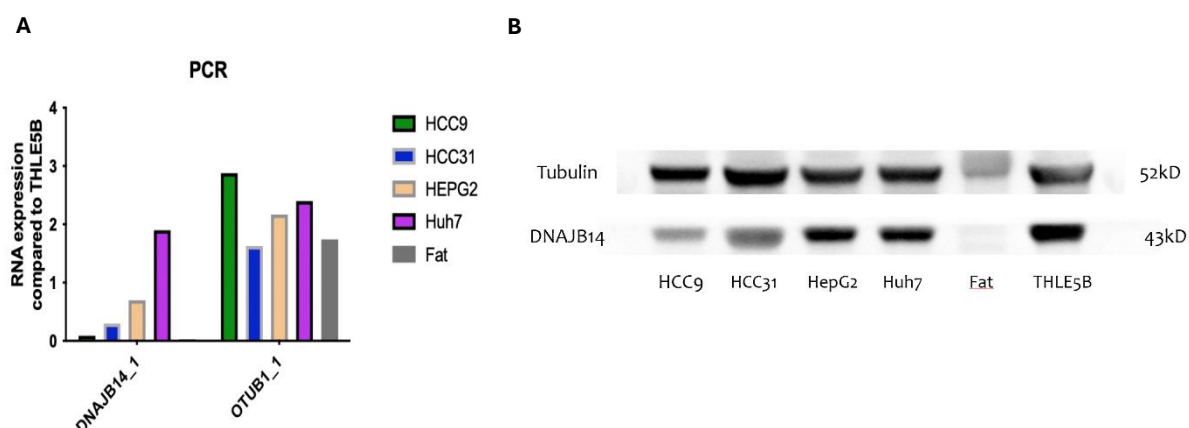
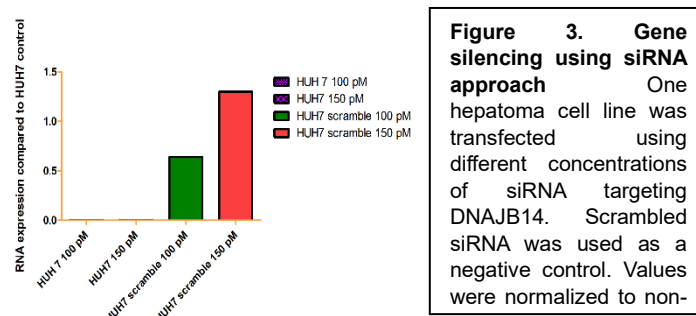


Figure 2. Gene and protein expression analysis in primary and established cell lines. A) qRT-PCR analysis of target genes in primary and established cell lines. Bars indicate relative gene expression relative to control (THLE-5B cell line). **B)** Western blot analysis of DNAJB14 in the cell lines. Tubulin was used as an loading control.

As two established cell lines, HepG2³ and Huh7, showed the highest levels of DNAJB14, they were further used for gene silencing to investigate the functional implications of this target. Here, siRNA gene silencing was performed and compared to non-transfected and scramble control cells. The results obtained clearly indicate that both concentrations of 100 and 150 pM effectively silenced the gene, with levels close to 0. Ongoing experiments include WB validation as well as functional effects by colony formation assay. As indicated in the plan above, all other functional experiments will be performed subsequently. Missing replicates will be included.



¹ As OTUB1 was not shown to be significant in the second report that was shared with us by UoS, we focused on DNAJB14. However, some of the experiments have been carried out in parallel or later for OTUB1.

² qRT-PCR for another biological replicates are ongoing

³ This cell lines will be further analyzed upon finishing the experiments with Huh7

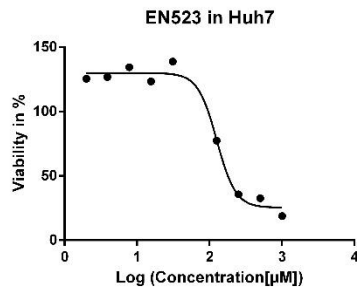


Figure 4. Dose-response analysis for OTUB1 inhibitor. Dose-response analysis using EN523 inhibitor specifically targeting OTUB1 in Huh7 cell line. Determined IC₅₀ value was 125μM.

Since commercial inhibitor for OTUB1 is available, we have performed initial determination of IC₅₀ values, which will later serve us for functional analyses (Figure 4)

References:

- 1 McGlynn KA, Petrick JL, El-Serag HB. Epidemiology of Hepatocellular Carcinoma. *Hepatol Baltim Md*. 2021;73 Suppl 1:4–13. doi: 10.1002/hep.31288
- 2 Frenette C, Gish RG. Hepatocellular carcinoma: molecular and genomic guideline for the clinician. *Clin Liver Dis*. 2011;15:307–21, vii–x. doi: 10.1016/j.cld.2011.03.010
- 3 Tejeda-Maldonado J, García-Juárez I, Aguirre-Valadez J, *et al*. Diagnosis and treatment of hepatocellular carcinoma: An update. *World J Hepatol*. 2015;7:362–76. doi: 10.4254/wjh.v7.i3.362
- 4 Marquardt JU, Andersen JB, Thorgeirsson SS. Functional and genetic deconstruction of the cellular origin in liver cancer. *Nat Rev Cancer*. 2015;15:653–67. doi: 10.1038/nrc4017
- 5 Farazi PA, DePinho RA. Hepatocellular carcinoma pathogenesis: from genes to environment. *Nat Rev Cancer*. 2006;6:674–87. doi: 10.1038/nrc1934
- 6 Das B, Toraman S. Deep transfer learning for automated liver cancer gene recognition using spectrogram images of digitized DNA sequences. *Biomed Signal Process Control*. 2022;72:103317. doi: 10.1016/j.bspc.2021.103317
- 7 Saldana M, VanderVorst K, Berg AL, *et al*. Otubain 1: a non-canonical deubiquitinase with an emerging role in cancer. *Endocr Relat Cancer*. 2019;26:R1–14. doi: 10.1530/ERC-18-0264
- 8 Mo L, Liu J, Yang Z, *et al*. DNAJB4 identified as a potential breast cancer marker: evidence from bioinformatics analysis and basic experiments. *Gland Surg*. 2020;9:1955–72. doi: 10.21037/gs-20-431